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673380044-4
673380290-9
673380031-3
673380067-2
673380512-7

QA/Test
Network Architects
IaC/DevOps
Network Architects
IaC/DevOps

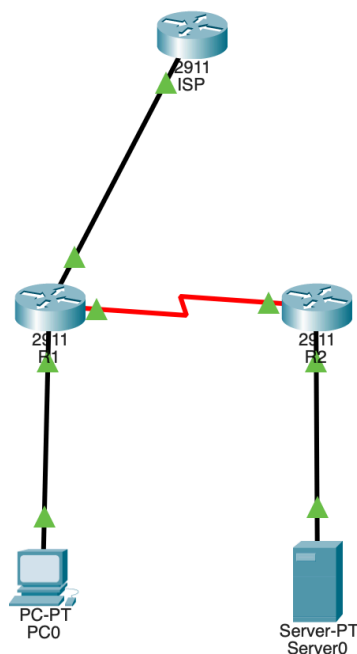
LAB 5 — Internet Edge + ISP Serial WAN + Week03 Microservices

LAB OBJECTIVE

Transform Week03 microservices into a **realistic enterprise deployment**:

- R1 = Internet Edge Router (NAT + DHCP WAN)
- Serial WAN = ISP Domain (100.10.10.0/30)
- R2 = Branch Router
- Microservices deployed on ServerA and ServerB
- Validate cross-site microservice access over routed WAN + NAT

LAB 5 TOPOLOGY (Reframed)





ADDRESS TABLE (FINAL)

R1

PhysicalConfigCLIAttributes

IOS Command Line Interface

```
Processor board ID FTX1240WS
3 Gigabit Ethernet interfaces
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

VLINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
VLINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
VLINE-5-CHANGED: Interface Serial0/3/0, changed state to up
VLINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

Router#enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#router rip
Router(config-router)#
Router(config-router)#end
Router#vlan database
Warning: It is recommended to configure VLAN from config mode,
as VLAN database mode is being deprecated. Please consult user
documentation for configuring VTP/VLAN in config mode.
Router(vlan)#
VSTP-5-CONFIG_1: Configured from console by console
Router(vlan)#
Router(vlan)#
Router#
VSTP-5-CONFIG_1: Configured from console by console

Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  200.0.0.2       YES manual  up          up
GigabitEthernet0/1  192.168.10.1    YES manual  up          up
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Serial0/3/0        100.10.10.1     YES manual  up          up
Serial0/3/1        unassigned      YES unset  administratively down down
Vlan1             unassigned      YES unset  administratively down down
Router#
```

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R2

PhysicalConfigCLIAttributes

IOS Command Line Interface

```
Cisco IOS Software, C2900 Software (C2900-UNIVERSALK9-M), Version 15.1(4)M3, RELEASE SOFTWARE (fc)
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 18-Jul-07 04:52 by pt_team
Image text-base: 0x2100F918, data-base: 0x24729040

This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wvl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISC02911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX15240WS
3 Gigabit Ethernet interfaces
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

VLINE-5-CHANGED: Interface Serial0/3/0, changed state to up
VLINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
VLINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

Router>
Router>
Router>
Router>show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  unassigned      YES unset  administratively down down
GigabitEthernet0/1  192.168.20.1    YES manual  up          up
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Serial0/3/0        100.10.10.2     YES manual  up          up
Serial0/3/1        unassigned      YES unset  administratively down down
Vlan1           unassigned      YES unset  administratively down down
Router>
```

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PC0

Physical Config **Desktop** Programming Attributes

IP Configuration [X]

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.10.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::200:CFF:FEC2:34AD

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

☐ Top

Server0

Physical Config Services **Desktop** Programming Attributes

IP Configuration [X]

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.20.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.20.1

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::201:96FF:FE7E:1EC1

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

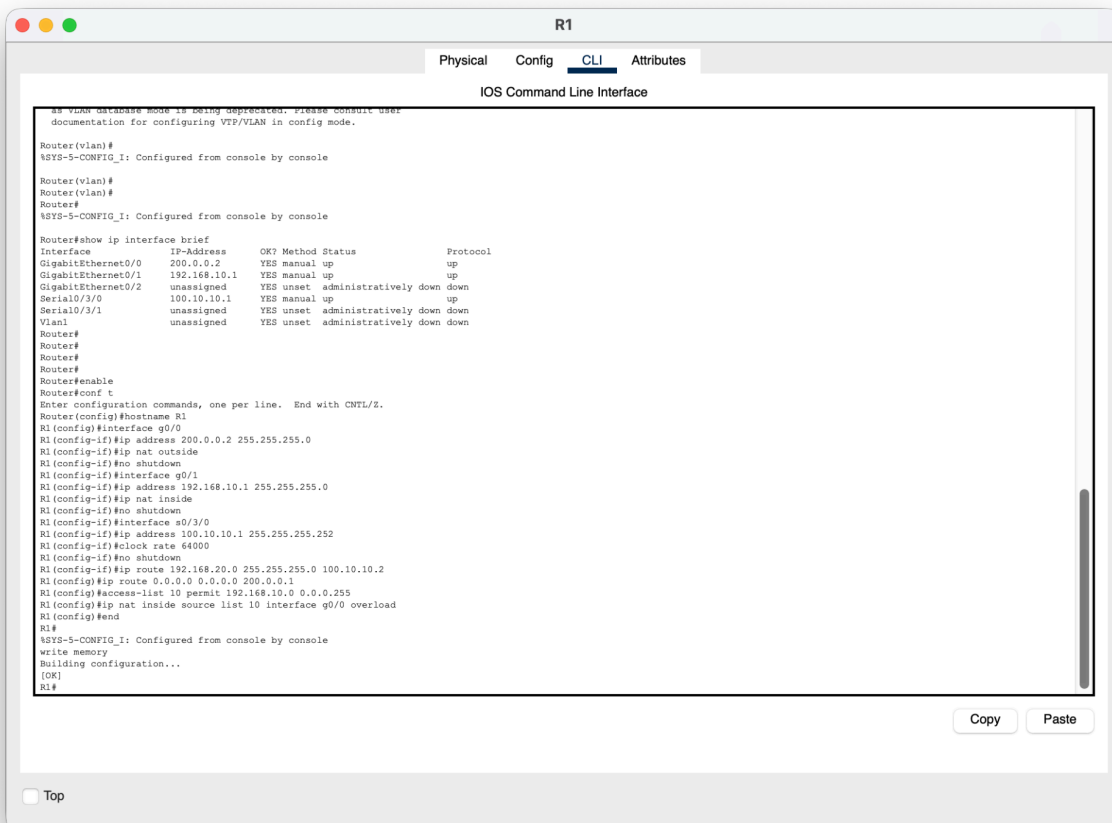
Password:

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Device	Interface	IP Address
R1	G0/0	200.0.0.2
R1	G0/1	192.168.10.1
R1	S0/3/0	100.10.10.1
R2	G0/1	192.168.20.1
R2	S0/3/0	100.10.10.2
ServerA	eth0	192.168.10.10
ServerB	eth0	192.168.20.10

PART 1 — CISCO CONFIGURATION (STRICT)

R1 — Internet Edge Router



```

R1
Physical Config CLI Attributes
IOS Command Line Interface

as vLAN database mode is being deprecated. Please consult user
documentation for configuring VTP/VLAN in config mode.

Router(vlan)#
%SYS-5-CONFIG_I: Configured from console by console

Router(vlan)#
Router(vlan)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface      IP-Address      OK? Method Status  Protocol
GigabitEthernet0/0  200.0.0.2      YES manual up       ip
GigabitEthernet0/1  192.168.10.1   YES manual up       ip
GigabitEthernet0/2  unassigned     YES unset  administratively down down
Serial0/3/0        100.10.10.1    YES manual up       ip
Serial0/3/1        unassigned     YES unset  administratively down down
Vlan1             unassigned     YES unset  administratively down down
Router#
Router#
Router#
Router#enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface g0/0
R1(config-if)#ip address 200.0.0.2 255.255.255.0
R1(config-if)#ip nat outside
R1(config-if)#no shutdown
R1(config-if)#interface g0/1
R1(config-if)#ip address 192.168.10.1 255.255.255.0
R1(config-if)#ip nat inside
R1(config-if)#no shutdown
R1(config-if)#interface s0/3/0
R1(config-if)#ip address 100.10.10.1 255.255.255.252
R1(config-if)#clock rate 64000
R1(config-if)#no shutdown
R1(config-if)#ip route 192.168.20.0 255.255.255.0 100.10.10.2
R1(config)#ip route 0.0.0.0 0.0.0.0 200.0.0.1
R1(config)#access-list 10 permit 192.168.10.0 0.0.0.255
R1(config)#ip nat inside source list 10 interface g0/0 overload
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
R1#

```

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R2 — Branch Router

R2

PhysicalConfigCLIAttributes

IOS Command Line Interface

```
Cisco IOS029117K9 (revision 1.0) With 491320K/32768K bytes of Memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

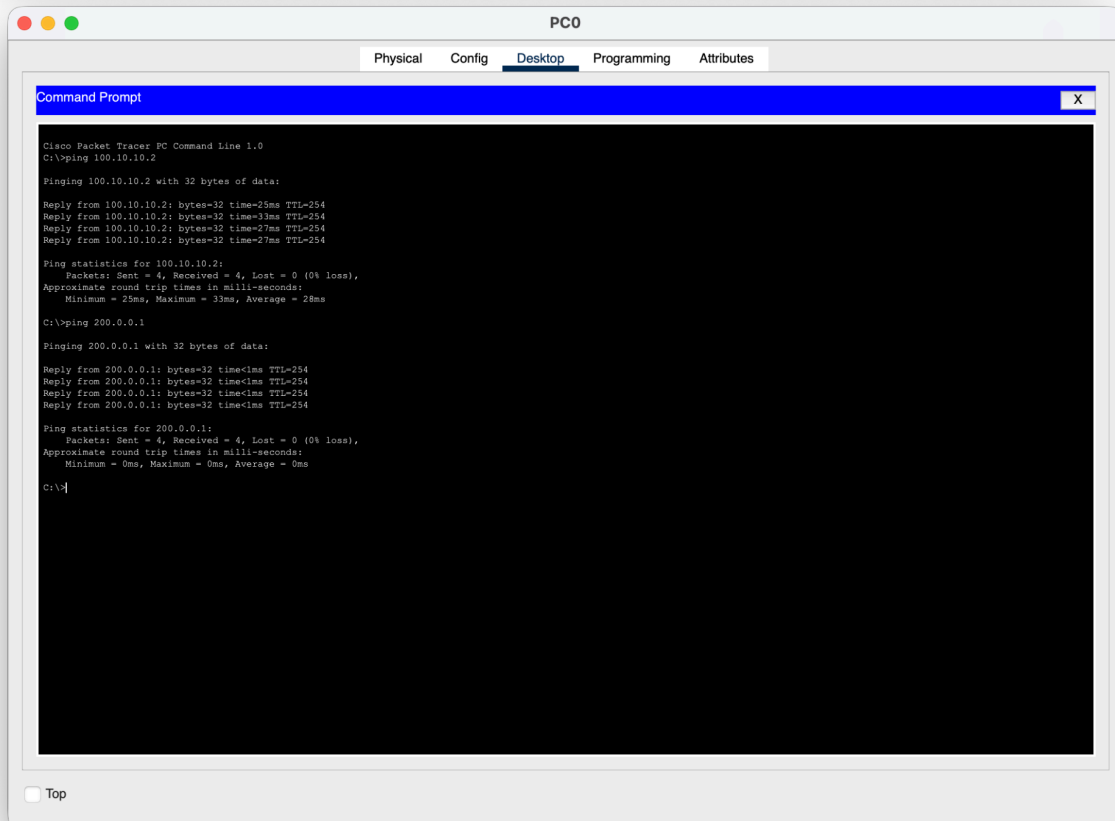
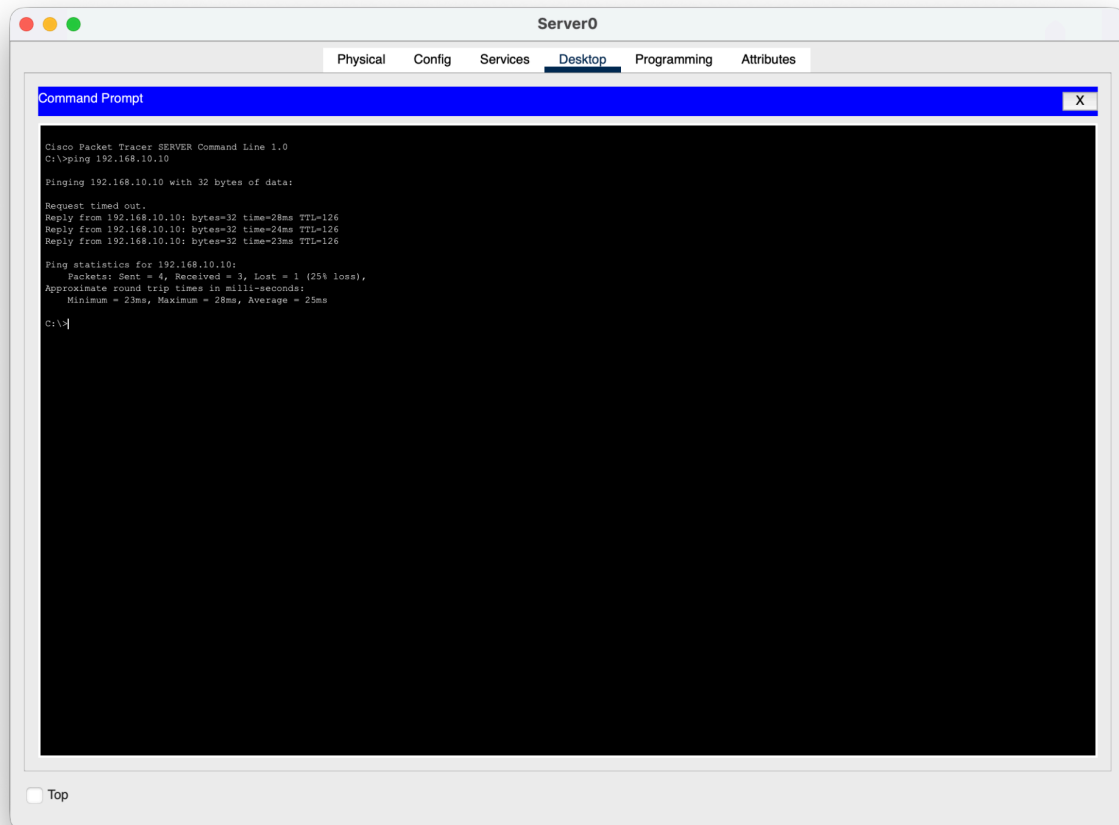
Router>
Router>
Router>
Router>show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0    unassigned      YES unset  administratively down down
GigabitEthernet0/1    192.168.20.1    YES manual  up          up
GigabitEthernet0/2    unassigned      YES unset  administratively down down
Serial0/3/0          100.10.10.2     YES manual  up          up
Serial0/3/1          unassigned      YES unset  administratively down down
Vlan1              unassigned      YES unset  administratively down down
Router>
Router>enable
Router>conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#hostname R2
R2(config)#interface g0/1
R2(config-if)#ip address 192.168.20.1 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#interface s0/3/0
R2(config-if)#ip address 100.10.10.2 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#ip route 192.168.10.0 255.255.255.0 100.10.10.1
R2(config)#ip route 0.0.0.0 0.0.0.0 100.10.10.1
R2(config)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
R2#
```

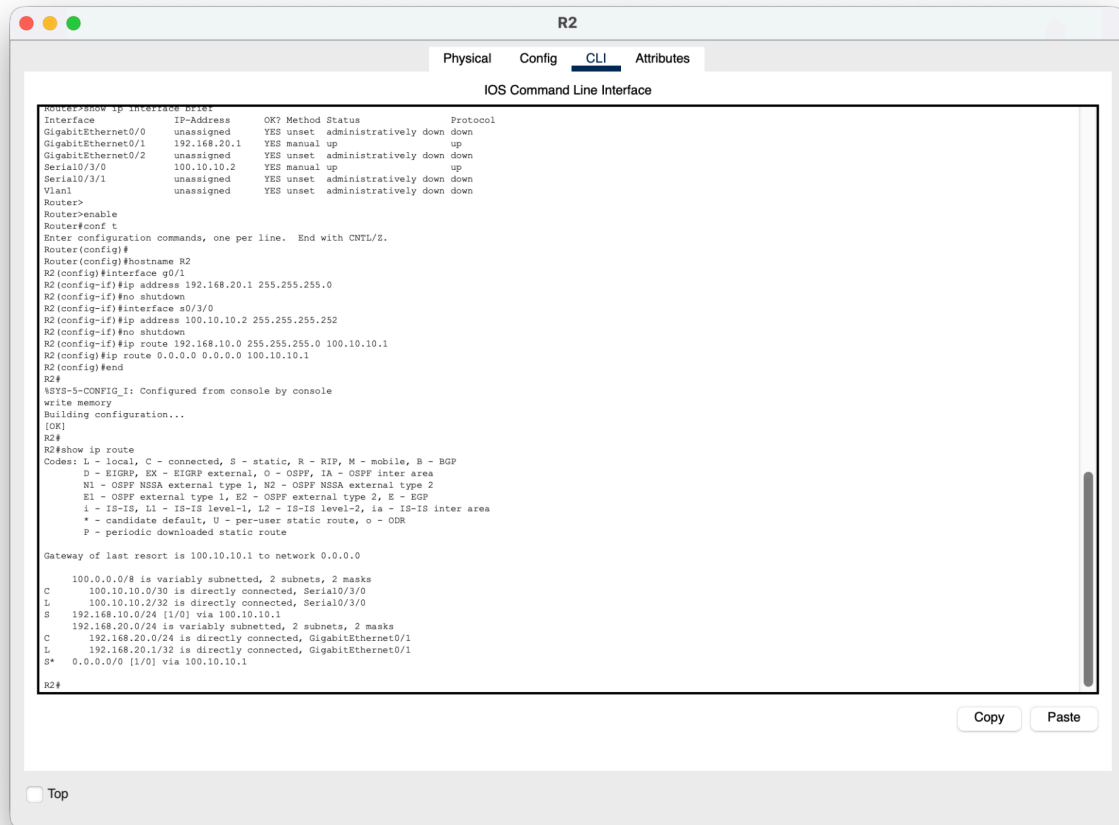
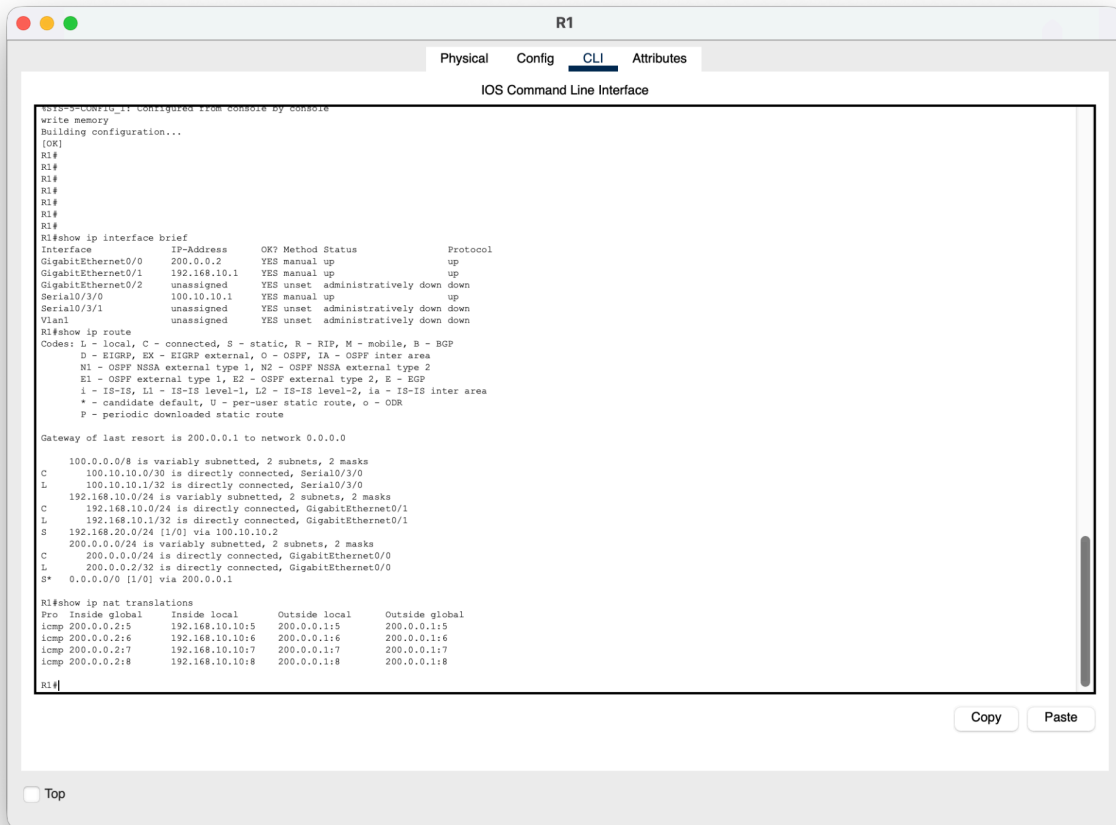
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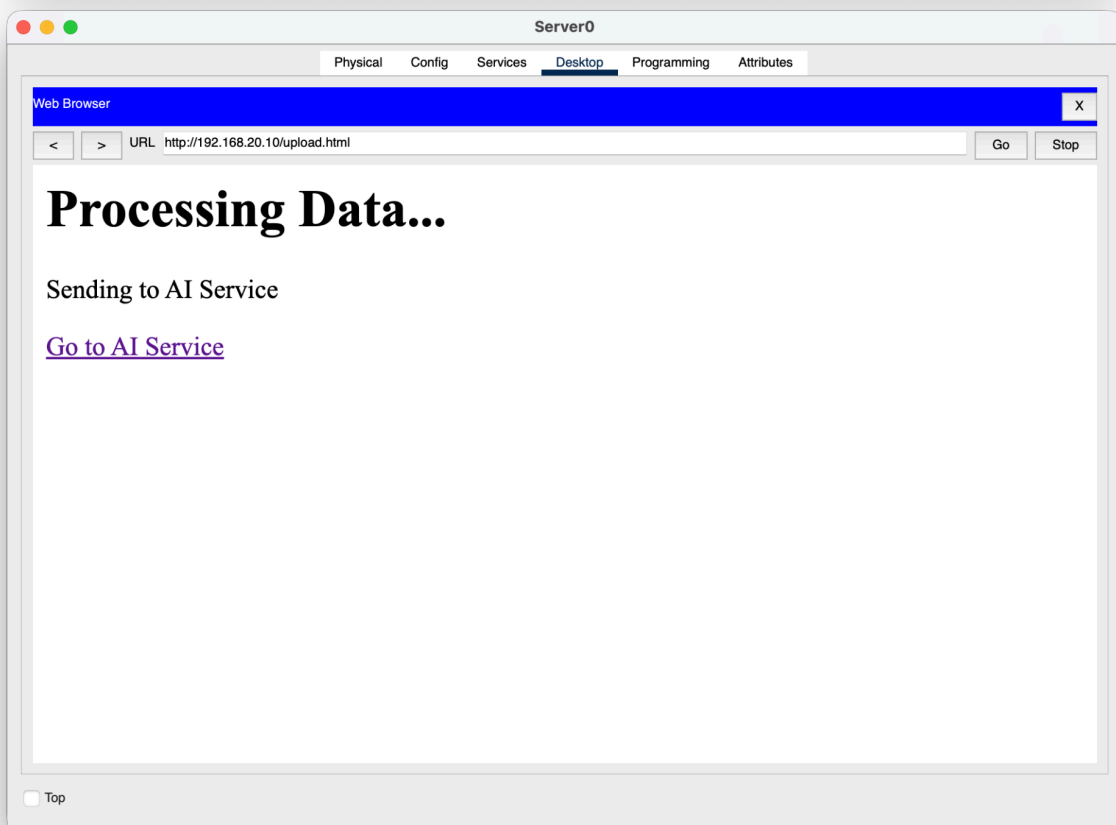
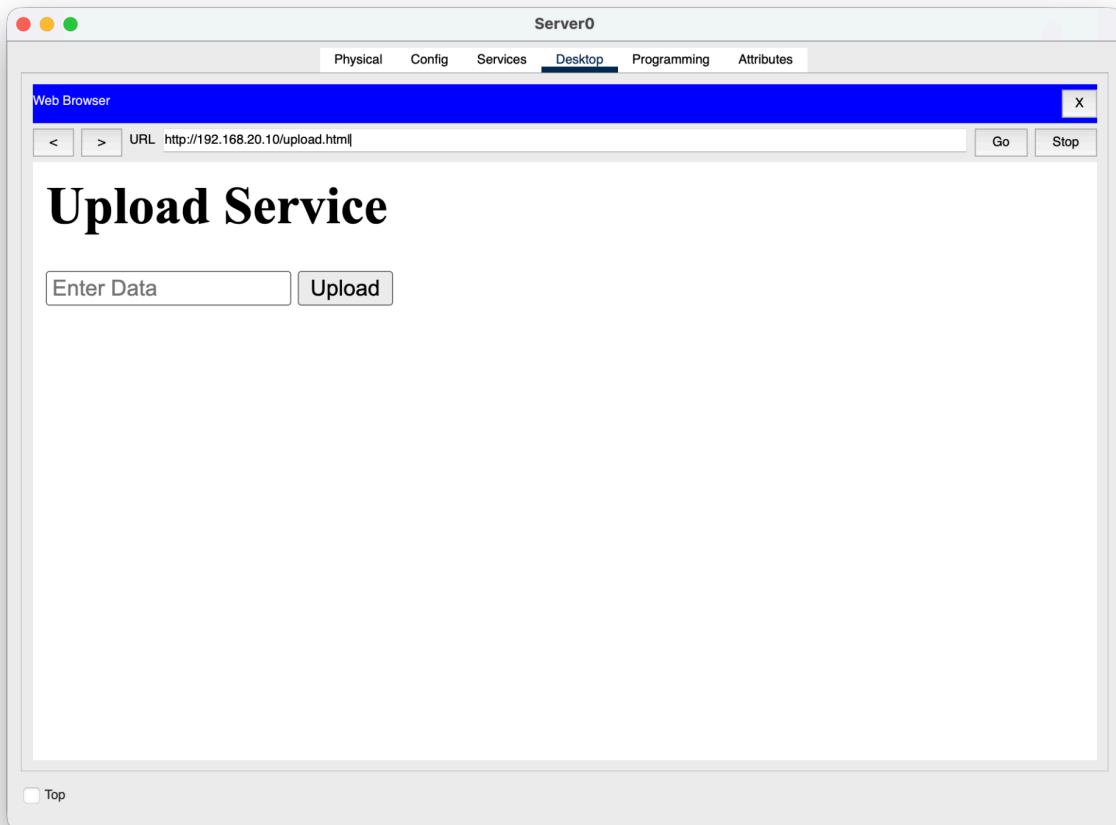
— Network Testing

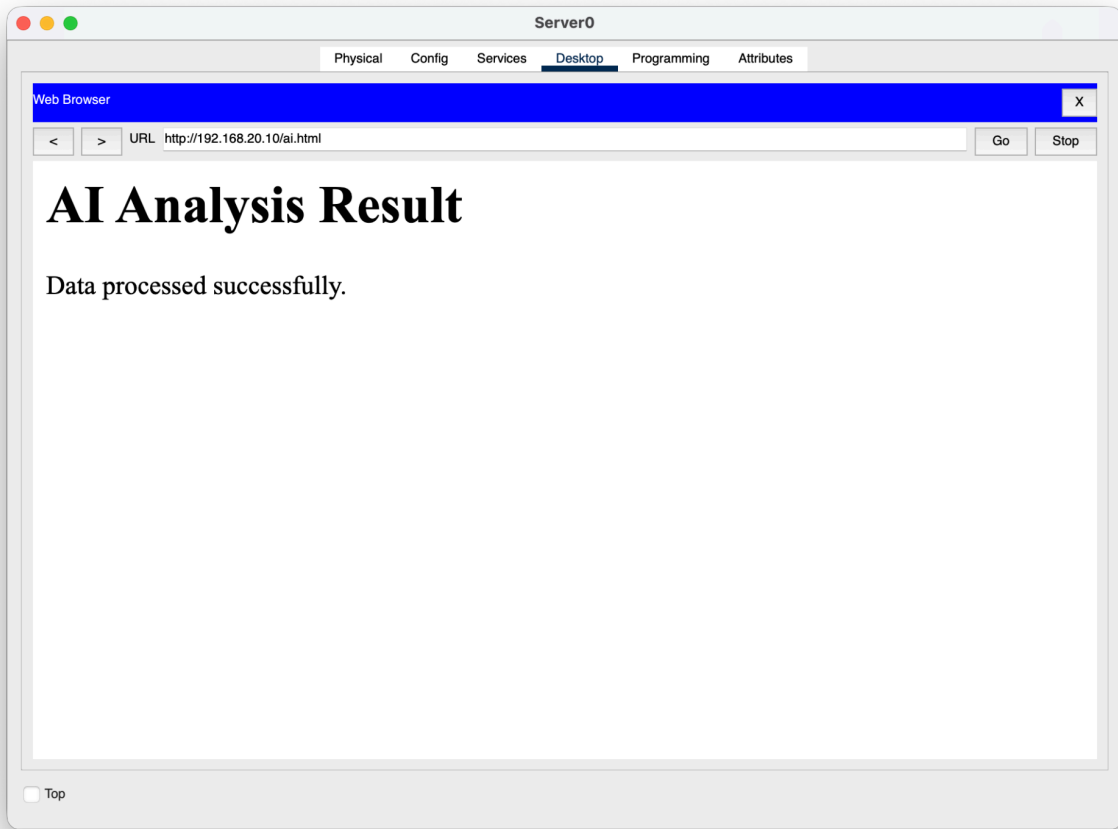






PART 2 — Deploy Microservices







PART 3 — WAN + INTERNET TESTING

Server0

Physical Config Services Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.10: bytes=32 time=28ms TTL=126
Reply from 192.168.10.10: bytes=32 time=24ms TTL=126
Reply from 192.168.10.10: bytes=32 time=23ms TTL=126

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 23ms, Maximum = 28ms, Average = 25ms
```

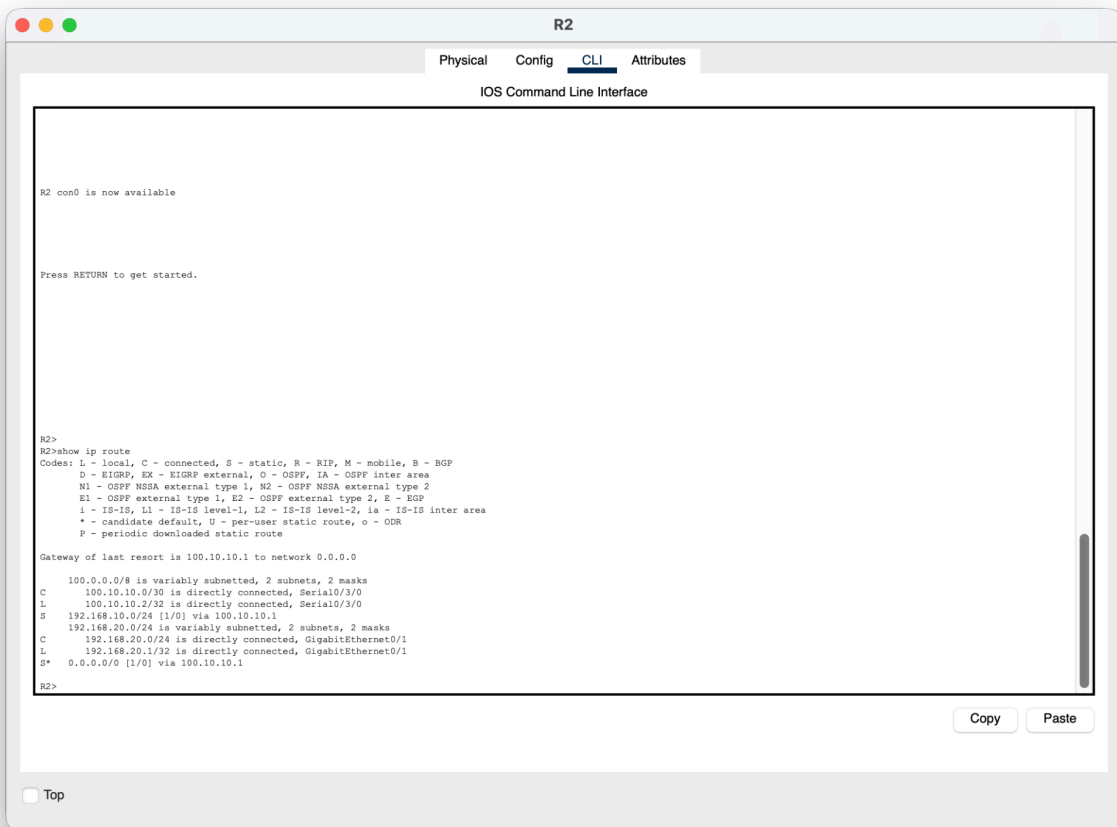
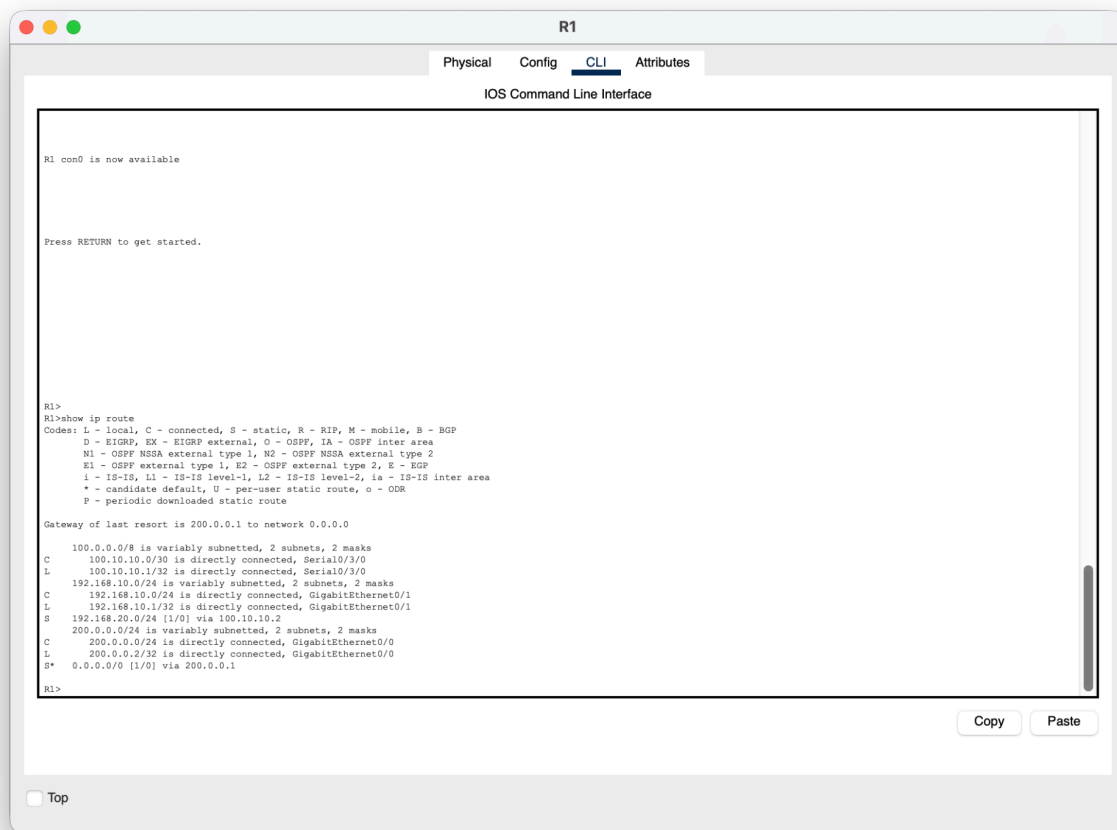
```
C:\>ping 100.10.10.2

Pinging 100.10.10.2 with 32 bytes of data:

Reply from 100.10.10.2: bytes=32 time<1ms TTL=255
Reply from 100.10.10.2: bytes=32 time<1ms TTL=255
Reply from 100.10.10.2: bytes=32 time<1ms TTL=255
Reply from 100.10.10.2: bytes=32 time<1ms TTL=255

Ping statistics for 100.10.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```





Engineering Reflection (Week03 Integration)

Engineering Reflection

This lab demonstrates the integration of networking concepts and distributed system architecture from Week03.

The network consists of two LAN segments connected through a serial WAN link using routers R1 and R2.

Router R1 acts as the Internet edge router and performs NAT translation, separating private and public networks.

The deployment of services across two servers simulates a distributed microservice architecture across multiple sites.



What Changed from Lab 4

Changes from Lab 4

Compared to Lab 4, this lab introduces a more realistic network architecture.

In Lab 4, the network used a simplified simulated internet network.

In Lab 5, the routers are connected through a serial WAN link representing an ISP network.

Additionally, NAT functionality is centralized at the edge router (R1), which better reflects real enterprise network design.



Architectural Significance

Architectural Significance

This architecture represents a simplified enterprise network design.

The system includes an edge router performing NAT, a WAN connection through an ISP network, and two separate LAN environments.

The deployment of services across two different network segments demonstrates how distributed systems operate across multiple locations.

This setup reflects real-world infrastructure used in cloud-edge and branch office environments.

